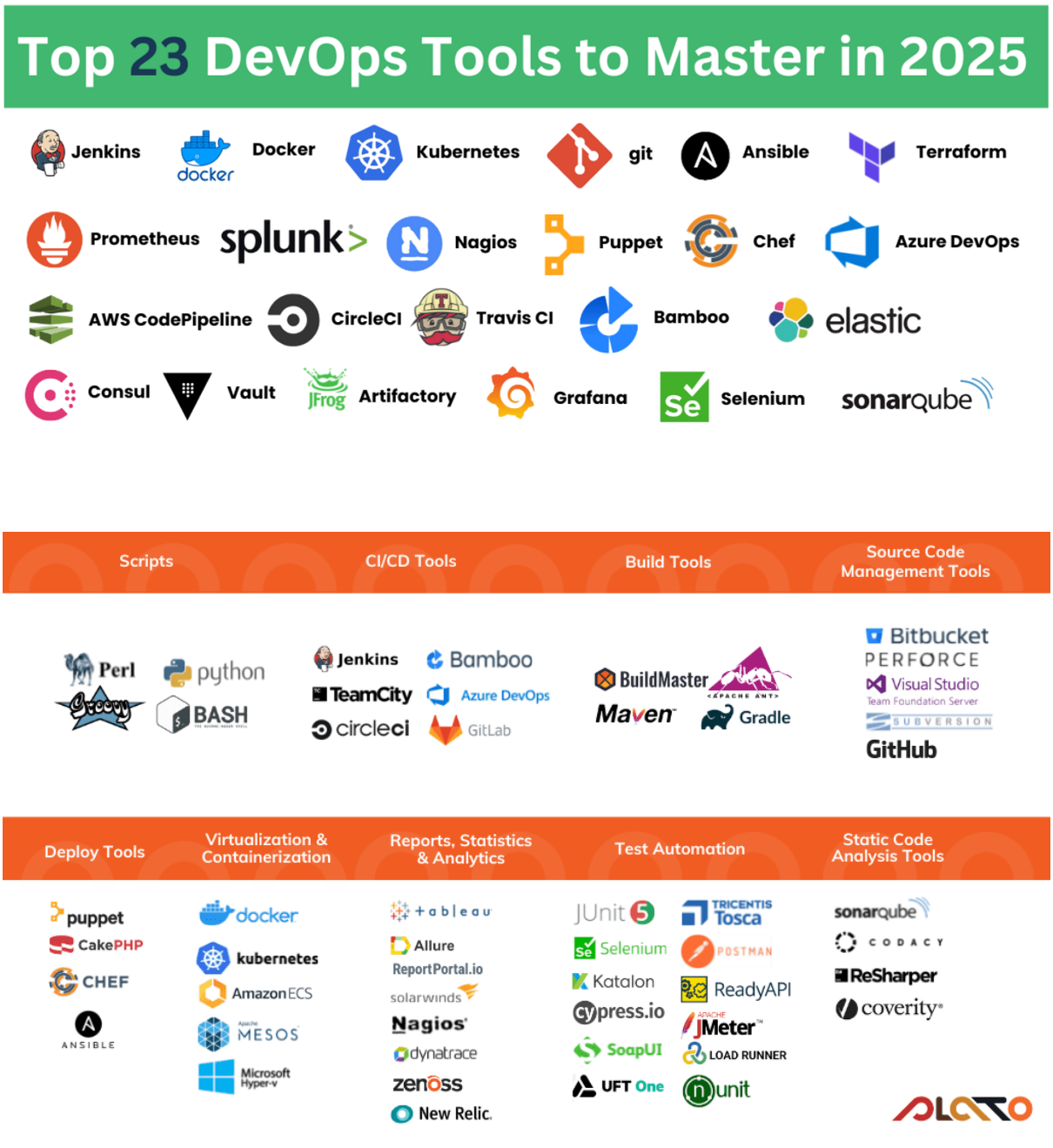
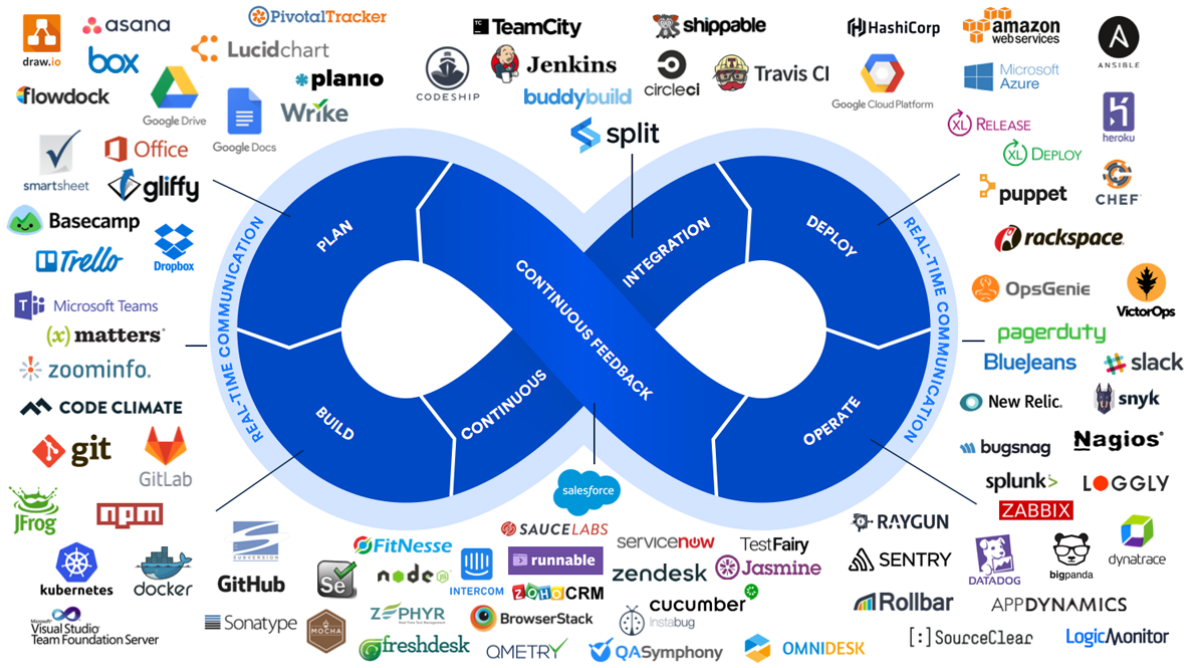
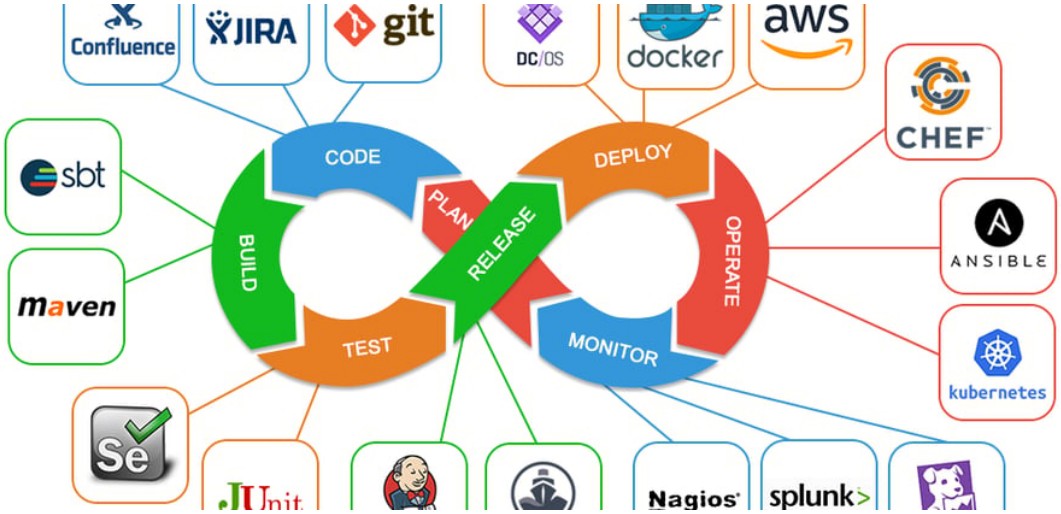
DevOps Tools



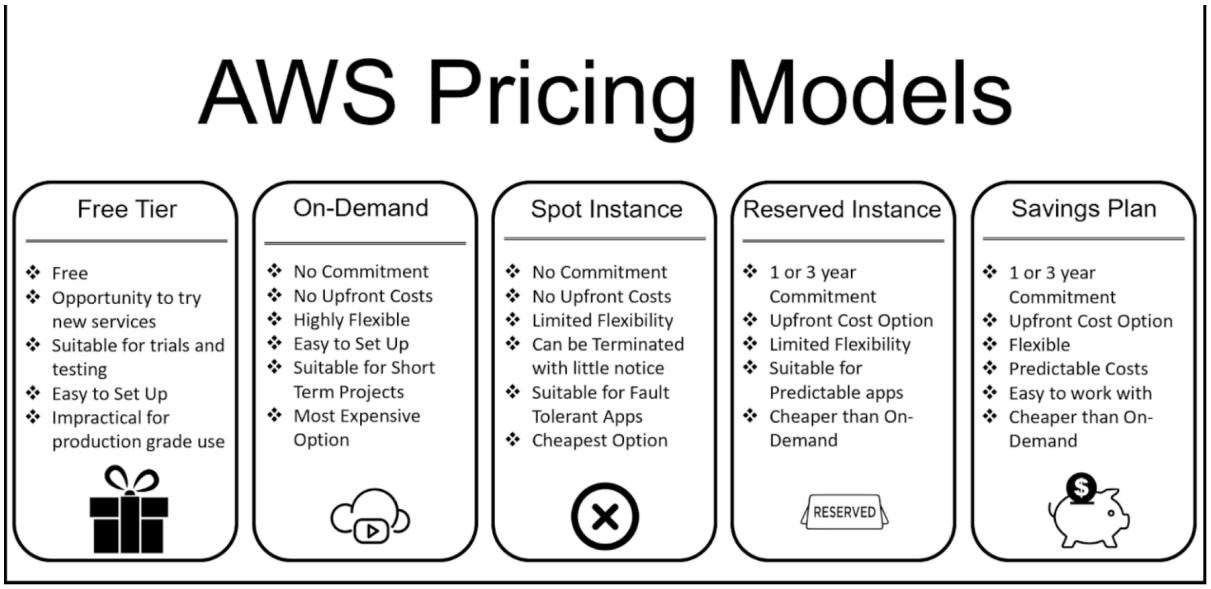
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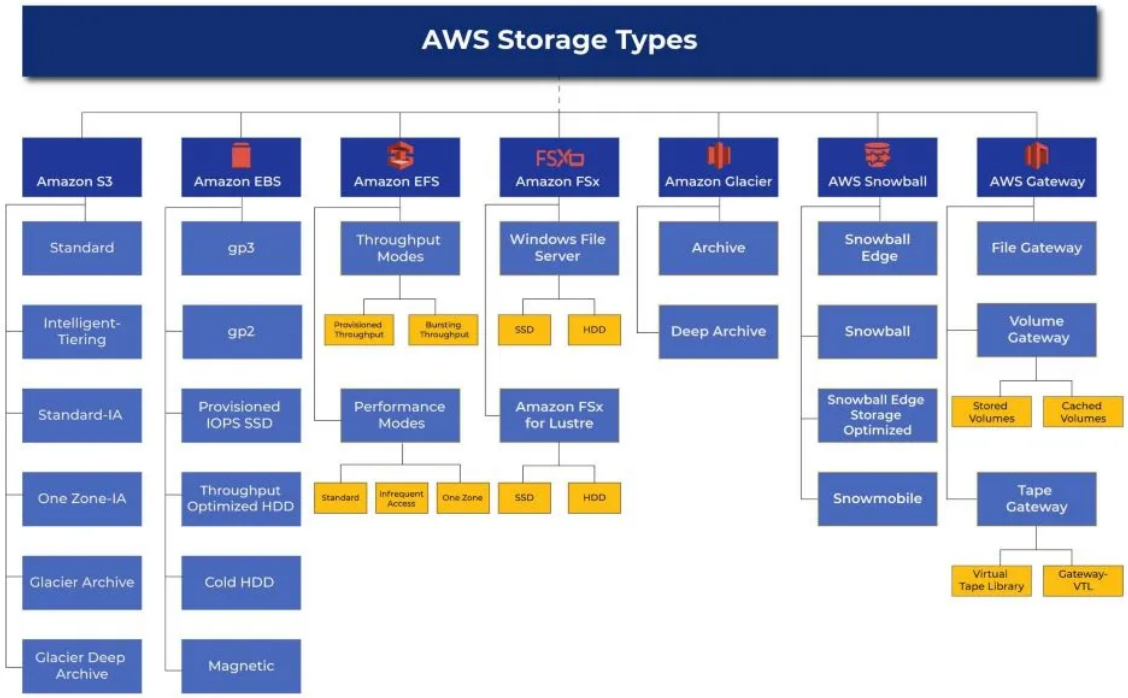


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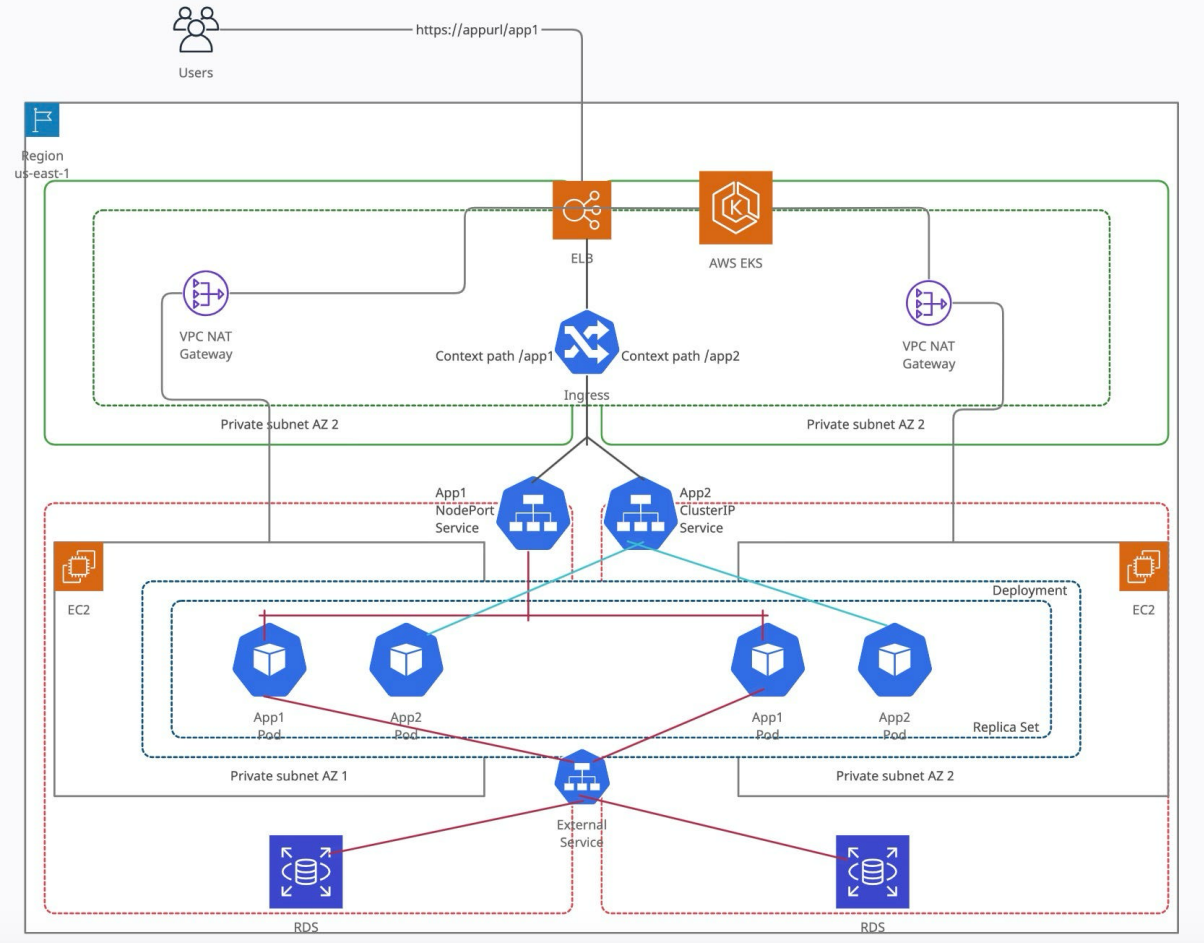
AWS PRICING MODEL



AWS Storage types



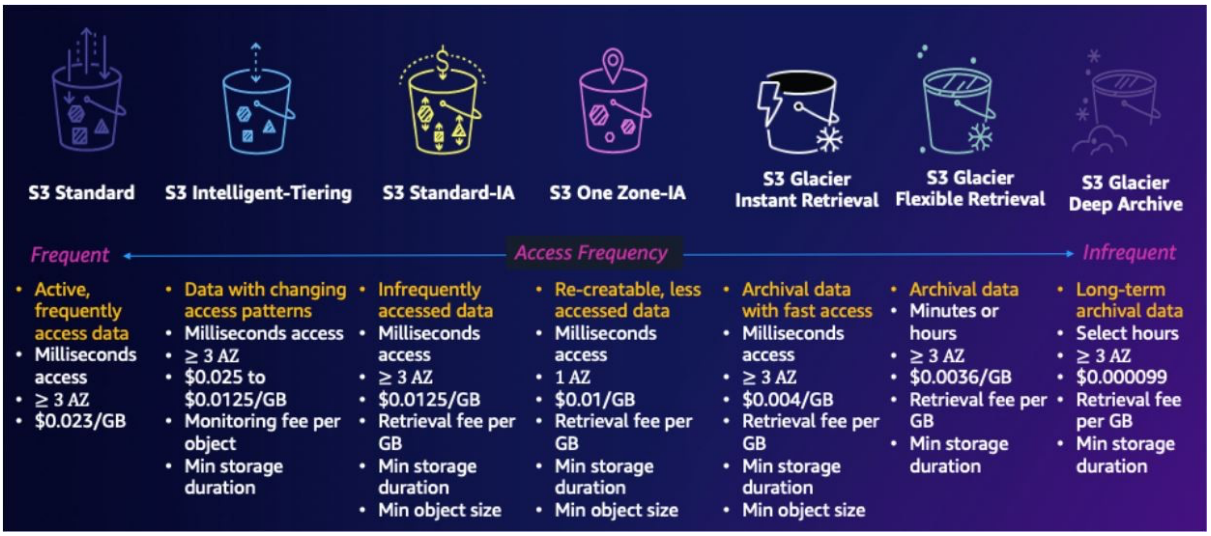
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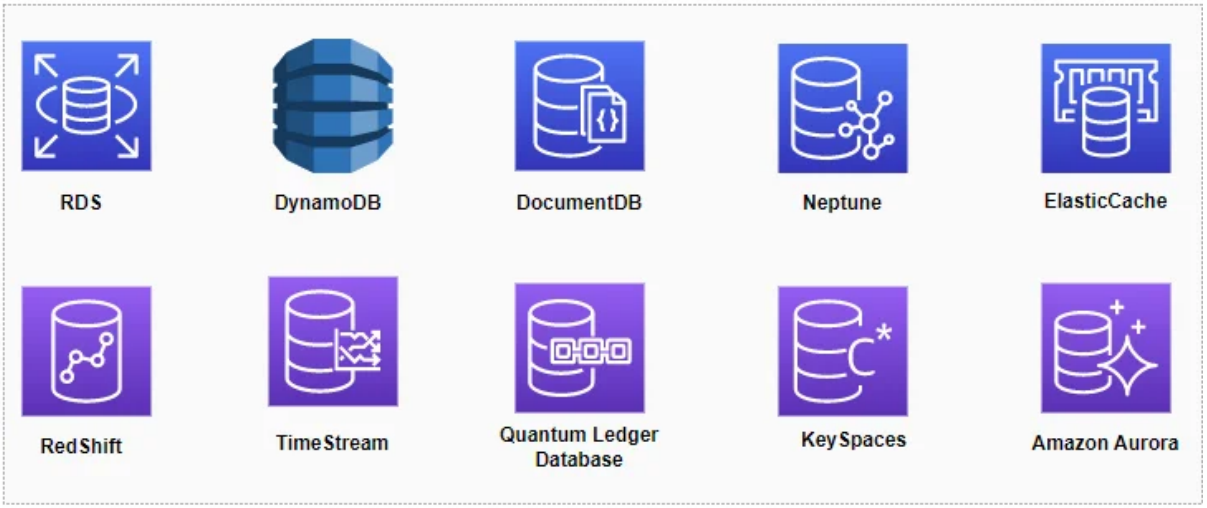
Database Type	AWS Service
Relational Database	*Amazon Arora* *Amazon RDS*
Key-Value pair	*Amazon DynamoDB*
In-Memory Cache	*Amazon Elastic Cache* *Amazon MemoryDB*
Document databases	*Amazon DocumentDB*
Wide Column Databases	*Amazon Keyspaces*
Graph Databases	*Amazon Neptune*
Time series (Realtime) databases	*Amazon Timestream*
Ledgers	*Amazon QLDB*

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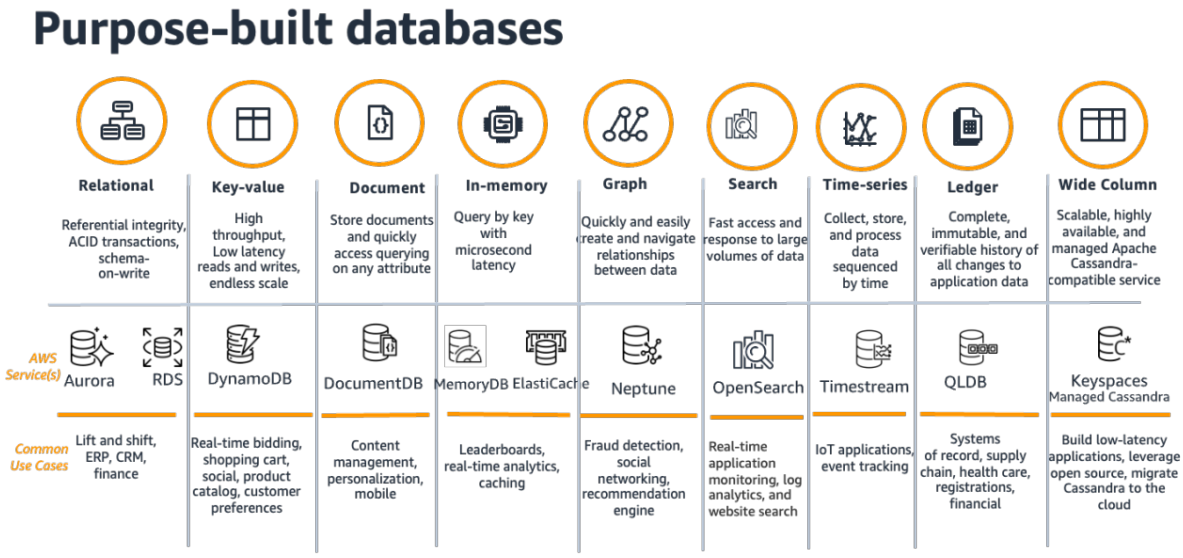
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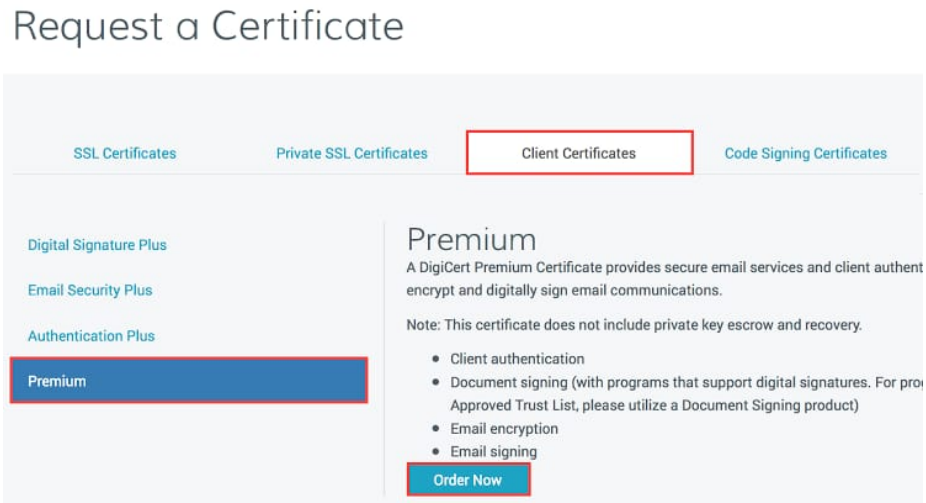
AWS Database Types



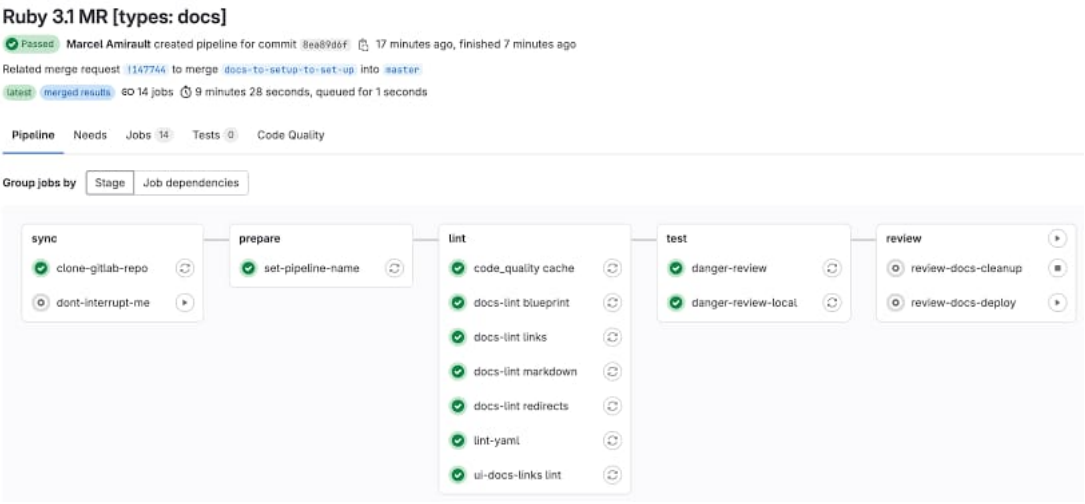
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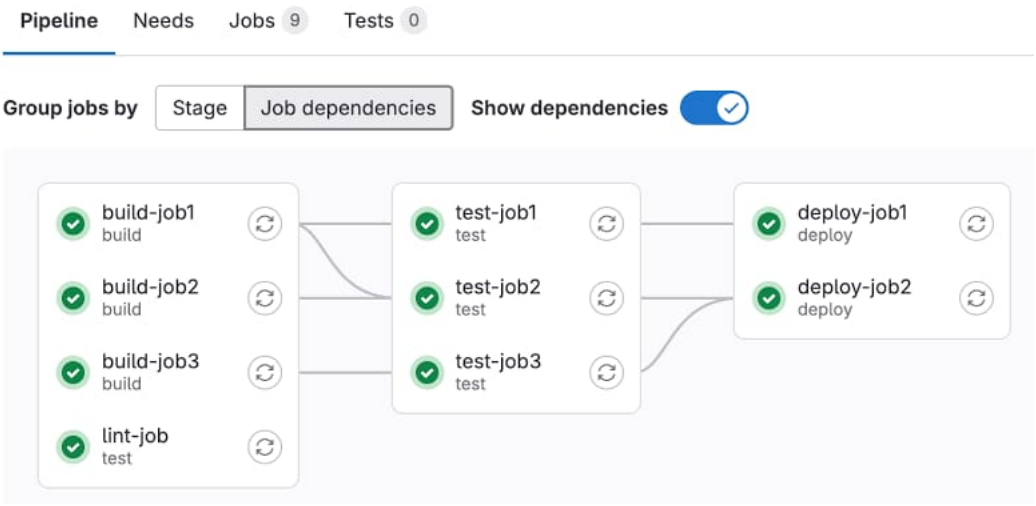
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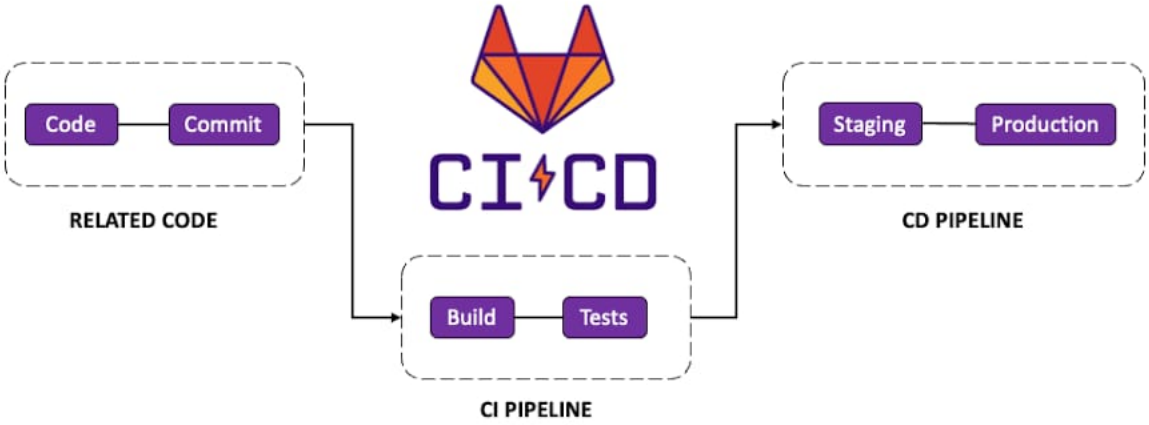


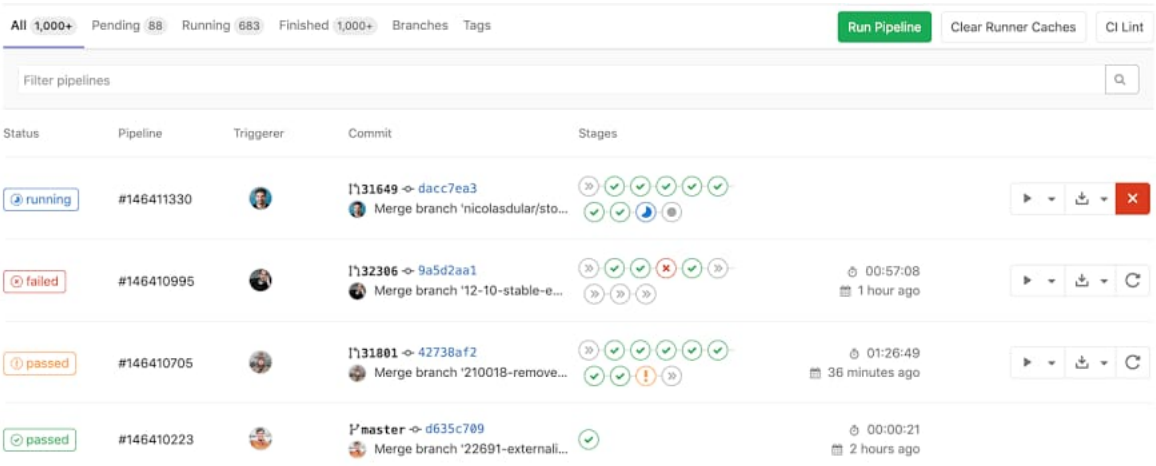
GIT LAB CI-CD



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Feature	Deployment	StatefulSet	ReplicaSet	DaemonSet
Purpose	Stateless applications	Stateful applications	Ensures fixed replicas of pods	Ensures a pod runs on all nodes
Pod Identity	Identical pods	Stable, unique pod identity	Identical pods	Identical pods, one on each node
Persistent Storage	No	Yes (via Persistent Volumes)	No	No
Pod Management	Manages pod replicas and updates	Manages stable, ordered deployment	Ensures a fixed number of replicas	Manages pods on each node
Scaling	Yes (with rolling updates)	Yes (ordered scaling)	Yes (but no rolling updates)	Yes (ensures one pod per node)
Rolling Updates	Yes (with zero downtime)	Yes (ordered updates)	No	No
Use Case	Stateless apps (web, microservices)	Stateful apps (databases, caches)	Pod replication (usually by Deployments)	Cluster-wide services (logging, monitoring)
Key Benefit	Easy management of stateless apps	Stable identity and persistent storage	Guarantees replicas without updates	Runs on every node in the cluster

NACL

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Security Groups	Network Access Control Lists - NACLs
Operates at the instance level	Operates at the subnet level
Applies to an instance only if it is associated with the instance	Applies to all instances deployed in the associated subnet (providing an additional layer of defense if security group rules are too permissive)
Supports allow rules only	Supports allow rules and deny rules
All rules are evaluated before deciding whether to allow traffic	Rules are evaluated in order, starting with the lowest numbered rule, when deciding whether to allow traffic
Stateful: Return traffic is allowed, regardless of the rules	Stateless: Return traffic must be explicitly allowed by rules
Multiple Security Groups can be associated with resources and a Security group can be be associated with multiple resources.	Multiple subnets can be bound with a single NACL, but one subnet can be bound with a single NACL only, at a time

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Network Load Balance vs Application Load Balancer

S.No	Network Load Balancer	Application Load Balancer
1	This is the distribution of traffic based on network variables, such as IP address and destination ports.	This is the distribution of requests based on multiple variables, from the network layer to the application layer.
2	It is layer 4 (TCP) and below and is not designed to take into consideration anything at the application layer such as content type, cookie data, custom headers, user location, or the application behavior.	This is a feature-filled L7 load balancer, HTTP, and HTTPS listeners only. Provides the ability to route HTTP and HTTPS traffic based upon rules, host-based, or path based.
3	The network load balancer works at layers 3 & 4 (network and transport layers).	Application Load Balancer (as the name implies) works at the Application Layer (Layer 7 of the OSI model).
4	Network load balancing cannot assure the availability of the application. This is because it bases its decisions solely on the network and TCP-layer variables and has no awareness of the application at all. Generally, a network load balancer will determine "availability" based on the ability of a server to respond to ICMP ping or to correctly complete the three-way TCP handshake.	An application load balancer goes much deeper and is capable of determining availability based on not only a successful HTTP GET of a particular page but also the verification that the content is as expected based on the input parameters.
5	A network load balancer will not differentiate between Application A and Application B when checking availability	An application load balancer will differentiate between the two applications by examining the application-layer data available to it.
6	NLB are better at handling huge spikes in traffic. Case in point, if you're running a retail site and are going to release a new, highly-demanded product, a NLB might be a better option to avoid a sudden traffic spike knocking out our website.	ALB examine the application layer protocol data from the request header. This examination takes more time but it enables the ALB to take a more informed decision about where to direct the traffic. This makes it slower in comparison to NLB.

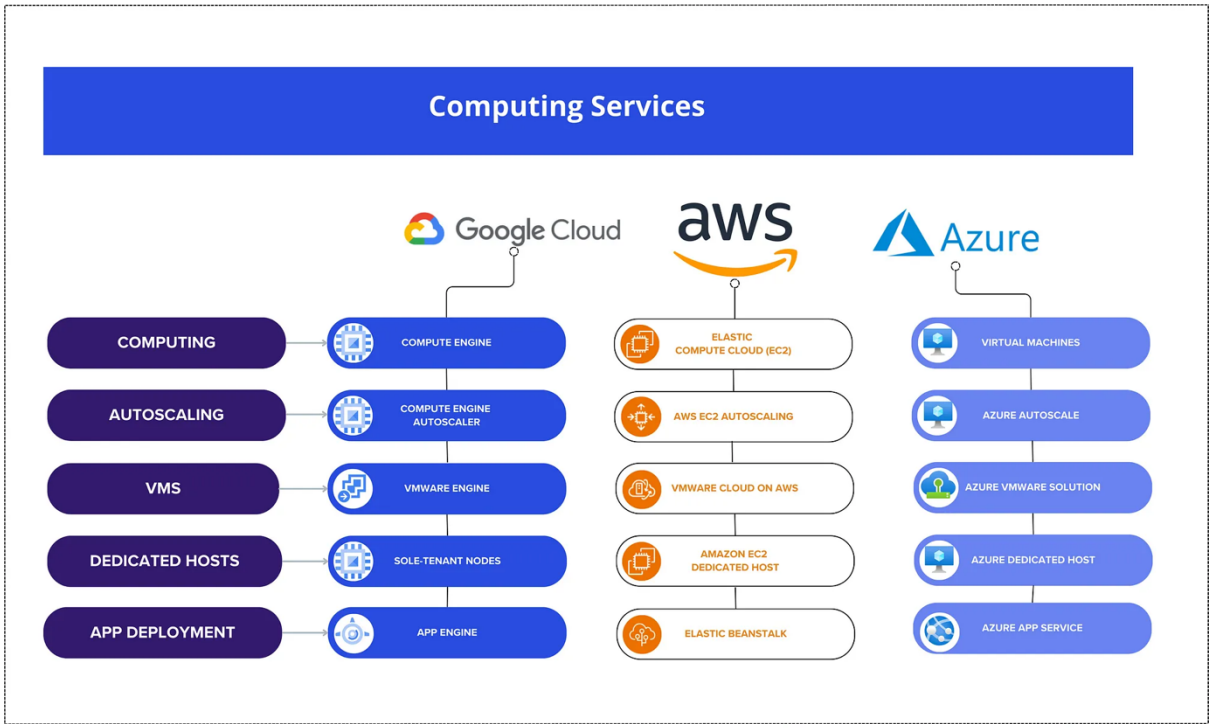
Load Balancer Type	Application Load Balancer (ALB)	Network Load Balancer (NLB)
Layer of Operation	Application Layer (Layer 7)	Transport Layer (Layer 4)
Ideal for	Distributing HTTP and HTTPS traffic	Handling TCP and UDP traffic, including databases
Content-Based Routing	Supported, enabling path and host-based routing	Not typically used for content-based routing
Advanced Features	SSL/TLS termination, WebSocket support	High-throughput, low-latency traffic handling

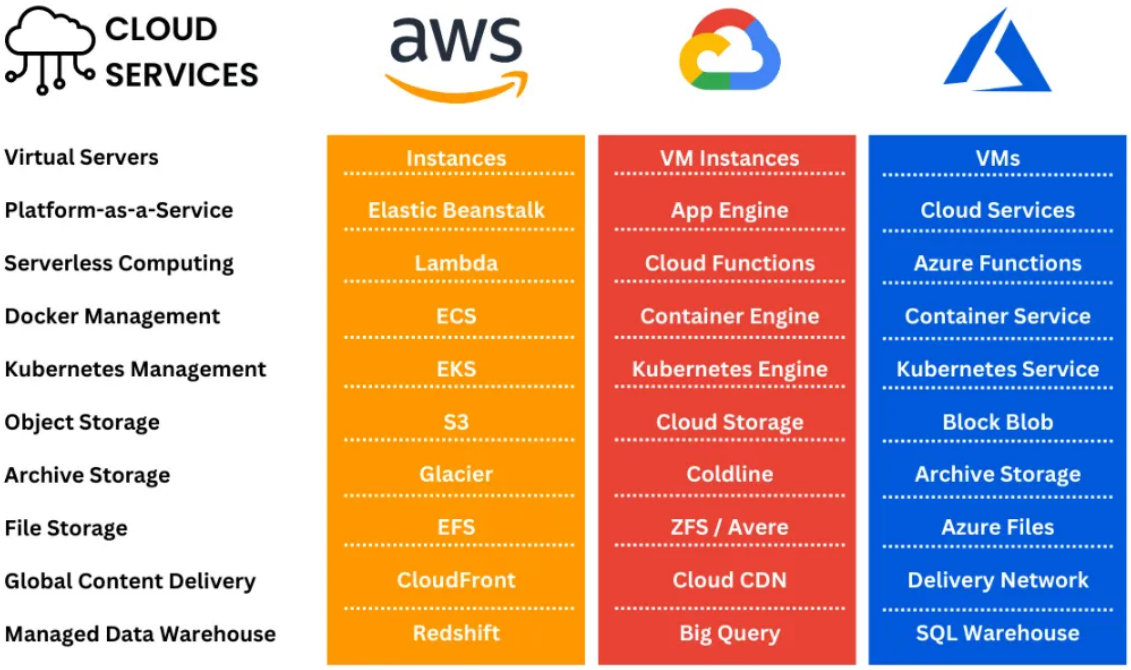
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Cloud Services Names



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Stateful VS Stateless Microservices

Feature	Stateful Microservices	Stateless Microservices
Session Management	Maintains user session state	No session state; each request is independent
Data Persistence	Stores data between requests	Does not store data; relies on external databases
Scalability	More complex; requires state synchronization	Easily scalable; can add more instances seamlessly
Fault Tolerance	Less fault tolerant; state can be lost on failure	More fault tolerant; failures do not affect others
Complexity	More complex due to state management	Simpler; no need for session management
Performance	Can be slower due to state handling	Generally faster due to stateless nature
Use Cases	E-commerce carts, gaming, financial transactions	APIs, content delivery, authentication
Deployment	Requires careful orchestration and management	Can be deployed more flexibly across environments
State Recovery	Needs strategies for state recovery	No need for state recovery as no state is stored
Consistency	May face challenges with data consistency	Consistency is easier as no state is shared

Aspect	Logs	Traces	Metrics
What It Is	Notes about specific events or errors.	Shows the path of a request across systems.	Numbers tracking system performance.
Purpose	Help troubleshoot problems.	Helps identify slowdowns and bottlenecks.	Tracks overall system health and performance.
When to Use	When you need details about what happened.	To see how a request moves through the system.	To monitor and alert on system performance.
Example	"Error: Cannot connect to database."	"Request went through Service A (1ms), Service B (50ms)."	"Request count: 10,000 per minute."

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Port Numbers & Services

Port Number	Protocol/Service	Explanation
80	HTTP	Default port for unencrypted web traffic.
443	HTTPS	Default port for secure web traffic over SSL/TLS.
22	SSH	Used for secure remote server access and management.
21	FTP	Used for unencrypted file transfer between servers.
3306	MySQL	Default port for MySQL database communication.
5432	PostgreSQL	Default port for PostgreSQL database communication.
2375	Docker (unencrypted)	Port for remote access to Docker API (unencrypted).
2376	Docker (encrypted)	Port for remote access to Docker API (encrypted).
8080	Jenkins	Default port for Jenkins CI/CD tool web interface.
9418	Git	Port for Git protocol, used for cloning repositories.
6443	Kubernetes (API Server)	Port for Kubernetes API server for cluster management.
9200	Elasticsearch	Default port for HTTP communication with Elasticsearch.
9090	Prometheus	Default port for Prometheus monitoring and querying interface.
3000	Grafana	Default port for Grafana monitoring dashboard web interface.
5666	Nagios	Port for remote monitoring communication with Nagios server.
5000	Docker Registry	Default port for private Docker registry.
8500	Consul	Default port for Consul service discovery web interface.
5900	VNC	Default port for Virtual Network Computing (remote desktop).
8081	Maven Repository	Default port for Maven repository access to artifacts.
9000	SonarQube	Default port for SonarQube web interface for code quality analysis.

Component	What It Does	Key Role in the Stack
Elasticsearch	Stores and indexes data, making it searchable	The search and data storage engine
Logstash	Collects, processes, and forwards data to Elasticsearch	The data pipeline that prepares and sends data
Kibana	Visualizes and displays the data from Elasticsearch	The tool for visualizing and interacting with data

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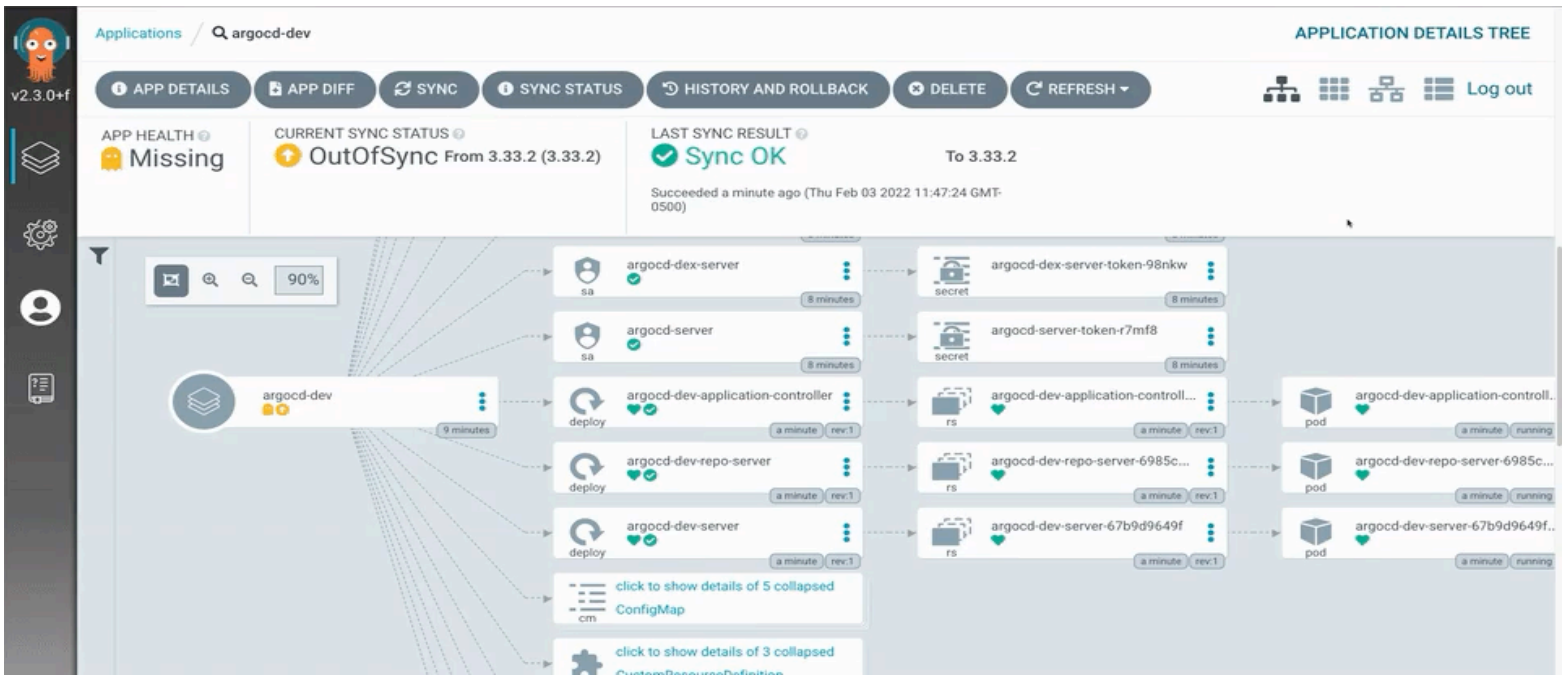
Concept	What It Is	When to Use	Example
IAM User	A person or service that interacts directly with AWS.	When an individual or service needs AWS access.	A developer with access to AWS Management Console or an API key.
IAM Group	A collection of IAM users for easier permission management.	To manage permissions for multiple users.	A "Developers" group that has specific permissions for development tasks.
IAM Role	A temporary identity with specific permissions, assumed by users or services.	For giving temporary access to resources.	An EC2 instance assuming a role to access an S3 bucket.
IAM Policy	A document defining specific permissions (allow or deny).	To define what users, groups, or roles can do.	A policy that allows read access to S3 but denies write access.

Feature	Datadog	ELK Stack
Log Management	Integrated log management and analytics in one platform.	Uses **Elasticsearch** to store and search logs, **Logstash** to collect and process, and **Kibana** to visualize.
Metrics	Built-in metrics monitoring for infrastructure and applications.	Can be integrated with other tools to collect metrics, but primarily focused on logs.
Application Monitoring	Provides Application Performance Monitoring (APM) to trace requests and visualize performance.	You can use **Elastic APM** (part of the Elastic Stack) but it's not as native as Datadog's APM solution.
Dashboards & Visualization	Interactive dashboards with built-in support for multiple data sources.	Kibana offers powerful, customizable visualizations and dashboards, but setup might require more effort.
Alerting	Native alerting with thresholds and anomaly detection.	Alerting can be configured with **Elasticsearch Watcher** or **Elastic Stack alerting features** (requires setup).
Cloud/On-Prem	Fully managed cloud service, no on-prem setup required.	Self-hosted or managed via **Elastic Cloud** (cloud-managed option).
Integrations	Pre-built integrations for various services (cloud platforms, databases, apps, etc.).	Requires more configuration for integrations, but has a wide range of supported plugins.
Ease of Setup	Quick setup with little configuration needed.	Can require significant setup, especially for larger or more complex environments.

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Factor	Datadog	ELK Stack
Ease of Setup	Quick and easy with minimal setup	Requires more setup, especially if self-hosted
Integration	Excellent cloud service integration	Requires more manual integration setup
Cost	Subscription-based, potentially higher cost at scale	Free if self-hosted, but with maintenance costs
Customization	Limited customization compared to self-hosted options	Highly customizable, especially with Elasticsearch
Support & Maintenance	Fully managed, less effort needed	Self-hosted requires more management
Scalability	Auto-scaling in the cloud	Must be managed manually if self-hosted

Argo CD



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